

10•10 11•11 12•12

TokoBli Campaign Evaluation & Product Page Analysis

Reinaldy Zulfananda Arkaan · 17 July 2026

[W2 JUN26 UBP] Reinaldy Zulfananda Arkaan Intermediate

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Business Overview & Disclaimer

About TokoBli

One of Indonesia's largest e-commerce platforms, selling across a wide range of product categories. It runs major sales on the "twin dates" (10/10, 11/11, 12/12) with heavy discounts and free shipping, when transaction volume can climb to roughly 10x a normal day.

Business goal: sharpen the campaign strategy, grow transactions, and get the most revenue out of a discount budget that stays efficient.

Project goal

- Evaluate the three previous campaign scenarios and turn the results into clear recommendations.
- Test whether a new product detail page design leads to higher-value transactions than the current one.

Dataset: 4,217 transactions (10 Oct – 12 Dec 2023), 14 columns: Transaction/Product/Customer ID, Status, Created At, Campaign, SKU, Price, QTY, Discount, Shipping Cost, Total Revenue, Category, Payment Method.

Methodology & Workflow

1

Data Understanding

Profile all 4,217 rows and check for missing values, duplicates, and wrong data types.

2

Data Cleaning

Drop columns over 50% empty, remove rows with missing values and duplicates, and fix data types.

3

Descriptive Analytics

Remove revenue outliers with the IQR method, then compute 12 summary statistics for QTY, Discount, and Revenue.

4

EDA

Compare the main metrics across campaigns and categories using SUMIFS, pivot tables, and charts.

5

Hypothesis Testing

Run a two-sample t-test on transaction value: the old product page vs the new one.

Tools: Google Sheets: QUARTILE, COUNTIF(S), SUMIFS, AVERAGE/MEDIAN/MODE, VAR, STDEV, SKEW, T.TEST, plus pivot tables and charts.

Problem Statement & Objective

Problem Statement

How can TokoBli maximize campaign revenue while maintaining efficient discount spending?

Objective

Identify the best-performing campaign by analyzing transactions, customers, revenue, quantity sold, and discount efficiency, then provide data-driven recommendations.

Data Cleaning: 4,217 → 4,209 Rows

Drop the 'Status' column	99.5% empty (4,195/4,217), well past the 50% threshold, so the column goes.	-1 column
Drop rows with missing values	3 rows had no Price/QTY/Discount (under 0.1%). We removed them rather than impute.	-3 rows
Remove duplicates	Data → Data cleanup → Remove duplicates	-5 rows
Fix data types	Created At → Date; Price, QTY, Discount, Revenue → Number	14 → 13 valid columns

4,217

rows before cleaning

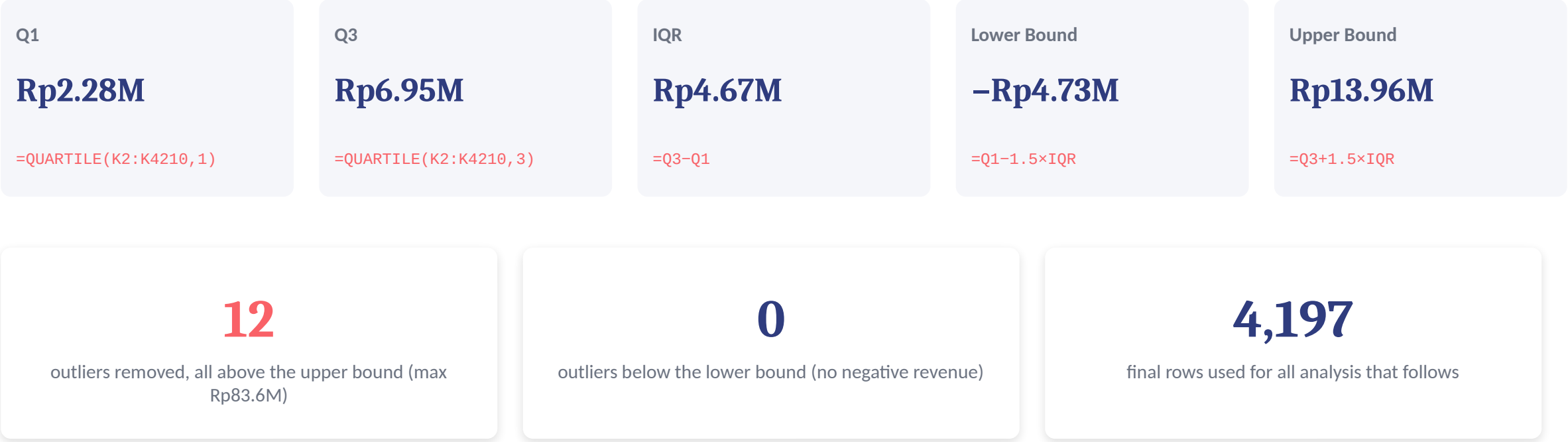
4,209

rows after cleaning

0

missing values & duplicates left

Outlier Removal: IQR Method on Total Revenue



Why IQR? Total Revenue is right-skewed, so the quartile approach holds up better than a z-score. Outliers were flagged with =IF(OR(K2<LowerBound, K2>UpperBound),"Outlier","OK") and then filtered out. Following the brief, we applied this to the whole dataset at once rather than per campaign. After removal, revenue skewness dropped to 0.39, close to symmetric.

Descriptive Statistics (n = 4,197)

Statistic	QTY	Discount (Rp)	Total Revenue (Rp)
Count	4,197	4,197	4,197
Min – Max	1 – 30	0 – 900,000	45,000 – 13,940,000
Mean	1.25	82,916	4,539,365
Median	1	0	4,200,000
Mode	1	0	7,200,000
Range	29	900,000	13,895,000
Q1 / Q3	1 / 1	0 / 0	2,280,000 / 6,900,000
Variance	0.92	3.57×10^{10}	6.98×10^{12}
Std Dev	0.96	188,969	2,642,368
Skewness	16.89	2.47	0.39

How to read it

- QTY is heavily skewed (16.9). Both median and mode are 1, so a typical order is just one item.
- Median and mode Discount are 0, meaning most transactions happen with no discount at all.
- After outliers were removed, revenue is nearly symmetric (skew 0.39), so the mean of Rp4.54M is a fair stand-in for a typical transaction.
- Revenue std dev is Rp2.64M, a fairly wide spread driven by price differences between categories.

Top 3 Business Insights

1

Order = 1 item

Median and mode QTY are both 1 (mean 1.25). Customers almost never buy more than one product per transaction, which leaves plenty of room for bundling, cross-sell, and minimum-spend offers to grow the basket.

2

Discounts aren't the main driver

More than half of all transactions happen with no discount (median and mode = 0). The average subsidy is only about Rp83K against Rp4.54M in revenue (roughly 1.8%), so the discount budget can be used more selectively without hurting sales.

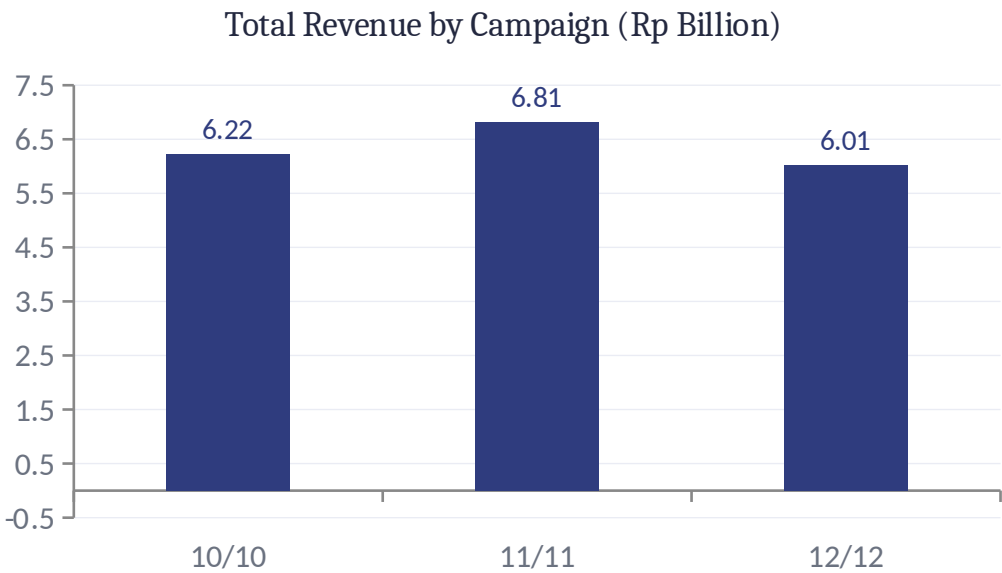
3

Revenue is healthy and steady

With the 12 outliers gone, revenue is close to symmetric (skew 0.39) and a typical transaction lands around Rp4.2–4.5M, a reliable baseline for setting next campaign's targets.

Comparing the Main Metrics Across Campaigns

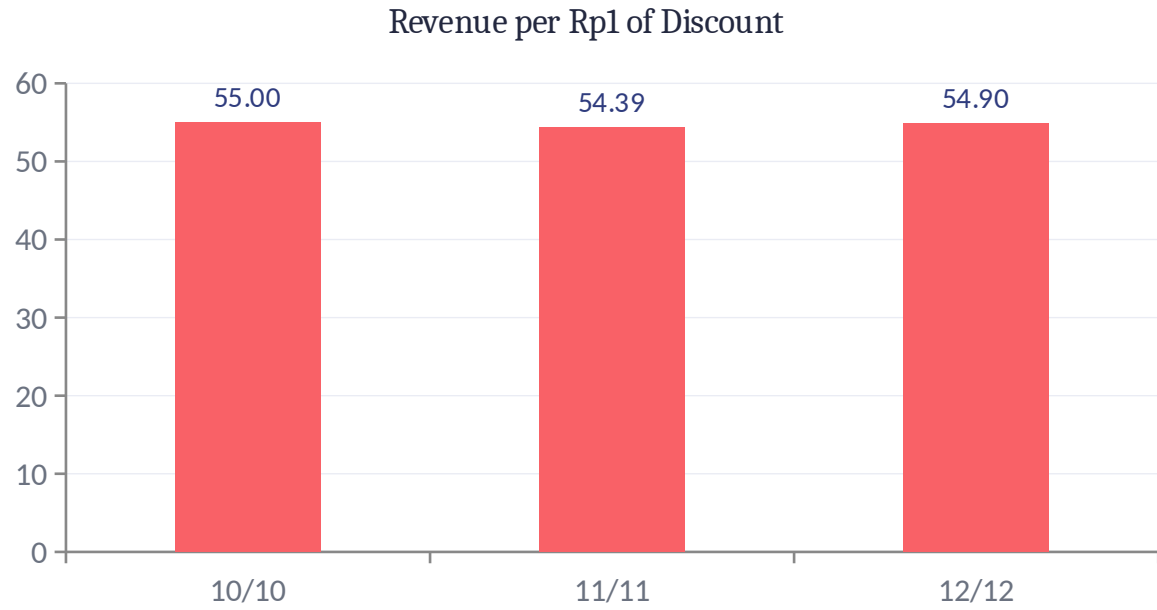
Metric	10/10	11/11	12/12
Total Transactions	1,399	1,398	1,400
Total Customers (unique)	890	879	855
Products Sold (QTY)	1,727	1,702	1,833
Total Revenue	Rp6.22B	Rp6.81B	Rp6.01B
Total Discount Budget	Rp113.2M	Rp125.3M	Rp109.5M



Insight

- Volume is almost identical. Transactions differ by just 2 across campaigns (1,398 to 1,400), so reach was basically the same.
- 11/11 wins on value: the highest revenue at Rp6.81B (+9.5% vs 10/10, +13.3% vs 12/12), even with the fewest customers. People spent more per transaction, not more often.
- 12/12 sold the most units (1,833) but earned the least revenue, so what sold there was cheaper on average.

Revenue Returned per Rp1 of Discount



How it's calculated: Ratio = Total Revenue ÷ Total Discount per campaign (SUMIFS). For 10/10:
 $6,224,149,500 \div 113,175,500 = 55.00$.

≈Rp54–55

revenue per Rp1 of discount, almost identical across all three campaigns

Insight

- Discount efficiency is flat: 55.00 vs 54.39 vs 54.90, a gap of only about 1%.
- Spending more on discounts didn't buy more efficiency. 11/11 spent the biggest budget (Rp125.3M) and earned the most revenue, but its ratio stayed the same.
- The takeaway: revenue grows in proportion to the subsidy. Budget size sets the size of the result, not how efficient it is.

The Standout Category: Men's Fashion

1,029

transactions, #1 of 15 categories

840

unique customers, #1

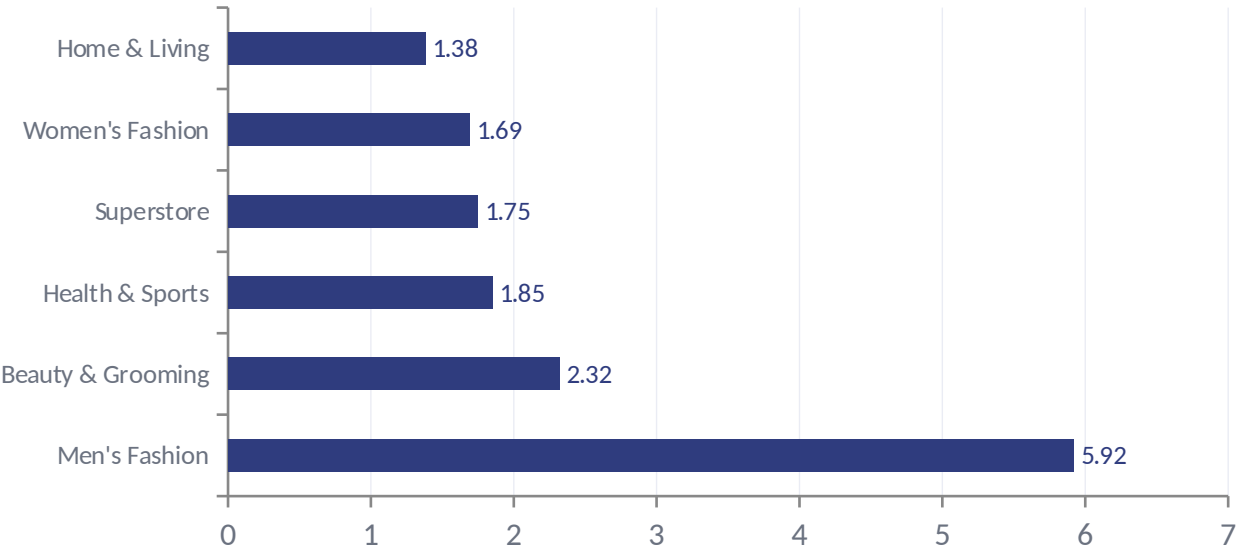
1,083

units sold, #1

Rp5.92B

revenue (31% of the total), #1

Top 6 Categories by Revenue (Rp Billion)



Leads every metric on a small subsidy

- Its discount spend was just Rp41.1M, well below Beauty & Grooming (Rp53.2M) and Superstore (Rp57.6M), whose revenue was less than half as much.
- Revenue-to-discount ratio of 144, nearly 3x the platform average of about 55.
- In short, Men's Fashion fits the brief exactly: the highest transactions, customers, units, and revenue on a relatively low discount budget.

Category Leaders by Campaign

Campaign	Most Transactions	Most Customers	Most Units	Highest Revenue
10/10	Beauty & Grooming (266)	Men's Fashion (208)	Beauty & Grooming (347)	Men's Fashion (Rp1.37B)
11/11	Men's Fashion (553)	Men's Fashion (457)	Men's Fashion (582)	Men's Fashion (Rp3.37B)
12/12	Health & Sports (252)	Men's Fashion (182)	Health & Sports (380)	Men's Fashion (Rp1.18B)

Insight

- Men's Fashion topped revenue in all three campaigns, and on 11/11 it swept every metric (553 transactions, 457 customers, 582 units, Rp3.37B, roughly half of that campaign's revenue).
- Beauty & Grooming drew a lot of transactions on 10/10 and Health & Sports on 12/12, but their transaction values were small. Busy doesn't always mean valuable.
- The jump for Men's Fashion on 11/11 (2 to 2.8x the other campaigns) points to a strong fit between the 11/11 scenario and this category, a pattern worth repeating.

Categories with the Highest Revenue/Discount Ratio

10/10

School & Education

384.9

Rp80.4M revenue against just Rp209K discount. A niche category with almost no subsidy.

11/11

Men's Fashion

171.5

Rp3.37B revenue against Rp19.6M discount. Efficient even at large scale.

12/12

Books

107.9

Almost tied with Men's Fashion (107.87). Both run about 2x the average.

Method note & takeaway: The ratio is computed per campaign × category as SUMIFS revenue ÷ SUMIFS discount; categories with zero discount (like Books on 10/10) are left out because the ratio is undefined. School & Education and Books look strong mainly because their subsidy is near zero, but they are small in scale. The one category that is efficient, large, and consistent (top-2 in every campaign) is Men's Fashion.

EDA Conclusions & Business Recommendations

11/11 is the best template · Men's Fashion is the growth engine

1 Replicate the 11/11 scenario

Same traffic as the other campaigns but the highest revenue (Rp6.81B). Use its mix of discounts and offers as the model for the next twin-date.

2 Put Men's Fashion first

Give it the main banner slot, deeper stock, and its own flash sales. It converts on a minimal subsidy (ratio 144 vs the 55 average).

3 Reallocate the discount budget

Shift subsidy away from low-ratio categories (Appliances 27, Superstore 30) toward high-ratio ones. Same total budget, more revenue.

4 Grow the basket size

A typical order is 1 item. Use bundles, conditional free-shipping tiers, and product recommendations to push QTY above 1.

Rough estimate: moving Rp30M of subsidy from a ratio of about 30 to about 140 could add roughly Rp3B in revenue per campaign.

A/B Test: Old Product Page vs New

Hypothesis for Old Product Page vs New Product Page

H₀: $\mu_A = \mu_B$
There is no significant difference in average transaction value between Group A and Group B.

H₁: $\mu_A \neq \mu_B$
There is a significant difference in average transaction value between Group A and Group B.

The test and its assumptions

Reject H₀
Since **p-value** (2.97×10^{-15}) < **0.05**, there is a statistically significant difference between Group A and Group B. The new product page (**Group B**) generated a higher average transaction value than the current product page (**Group A**).

Business Recommendation :

Implement the new product page (Group B) as the default version because it significantly increases the average transaction value. Continue monitoring post-launch performance through additional A/B testing to ensure consistent business impact.

Test Summary:

Metric	Group A	Group B
Mean Transaction Value	Rp745,890	Rp830,460
Standard Deviation	Rp149,864	Rp181,903
Statistical Test	Welch Two-Sample T-Test	
p-value	2.97×10^{-15}	
Significance Level	$\alpha = 0.05$	

Test Results & Recommendation (OBIPR Framework)

Observation	Business Impact	Isolation	Prioritization	Recommendation
Replicate the 11/11 campaign strategy	11/11 generated the highest revenue (Rp6.81B) despite having a similar number of transactions as the other campaigns.	The campaign strategy, promotion mix, and pricing approach used during 11/11 delivered the highest average transaction value.	H1 : Highest Priority Replicating the most successful campaign can produce the fastest business impact.	Adopt the 11/11 campaign strategy as the benchmark for future twin-date campaigns while maintaining a similar discount efficiency.
Optimize discount budget allocation	Several categories received relatively high discounts but generated lower revenue efficiency.	Low Revenue-to-Discount Ratio categories such as Superstore and Appliances reduced the overall return on the discount budget.	H2 : High Priority Budget optimization improves ROI without increasing marketing spendin	Reallocate discount budgets from low-efficiency categories to high-performing categories such as Men's Fashion to maximize revenue.
Increase average basket size	The average customer purchased only 1.25 items , indicating limited cross-selling opportunities.	Most transactions contain only one product, limiting revenue growth per transaction.	H3 : Medium Priority. Requires marketing and product initiatives but offers sustainable long-term growth.	Introduce product bundles, cross-selling recommendations, and minimum-spend free shipping promotions to encourage customers to purchase more items per transaction.

Summary: Getting the Most Revenue from the Next Campaign

Rp6.81B

11/11 revenue, the best scenario. Make it the template for the next campaign.

144 : 1

Men's Fashion revenue/discount ratio. Prioritize it and stock it deeper.

≈55

The flat platform ratio. Grow revenue by reallocating budget, not by spending more overall.

QTY = 1

The typical basket. Use bundles and free-shipping tiers to lift the value per order.

Thank You

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URL Spreadsheet :

https://docs.google.com/spreadsheets/d/1ec_Z0g4CwWygPyOb5oMyZOdFyFHPgfkLve325YZ1MHM/edit?usp=sharing